

AI-assisted soil intelligence for Ethiopian agriculture

From soil diagnosis to verified restoration: laboratory-aware AI, scientist review, controlled amendment matching, and field-trial traceability.

EAll Startup Support Program — Round 2

Maximum 7-slide proposal

The ask

HPC + soil-data partnerships + lab validation + controlled field pilots

Butterfly Tech / GerahtAI



Soil degradation is becoming an intelligence problem, not only an input problem

Farmers, institutions, and public programs need soil-specific evidence before they can restore productivity responsibly.

Why this matters now

~60%

of workforce in agriculture

~33%

of GDP from agriculture

>85%

of land reported degraded

15.5M

quintals fertilizer imported

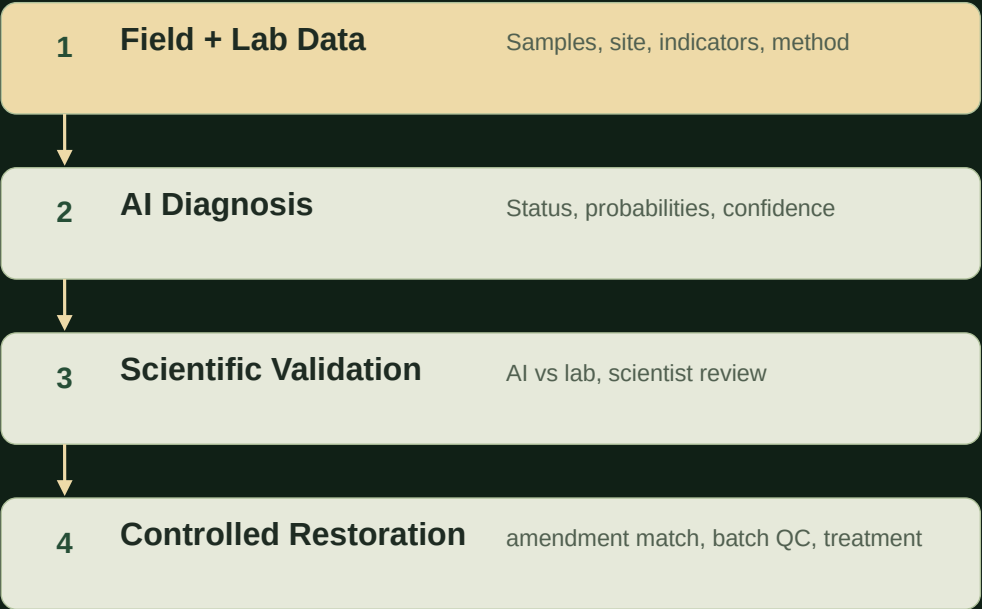
The opportunity is to reduce blind input use by creating trusted local soil intelligence — then linking diagnosis to safe, soil-specific restoration actions.

Unserved gap

Soil testing, fertilizer advice, AI tools, compost/biochar, and field programs usually operate separately. GerahtAI connects them into one validation-aware system.

A controlled soil-intelligence workflow — not a black-box recommendation app

GerahtAI turns field and laboratory evidence into decision-support outputs that can be reviewed, corrected, approved, and traced.

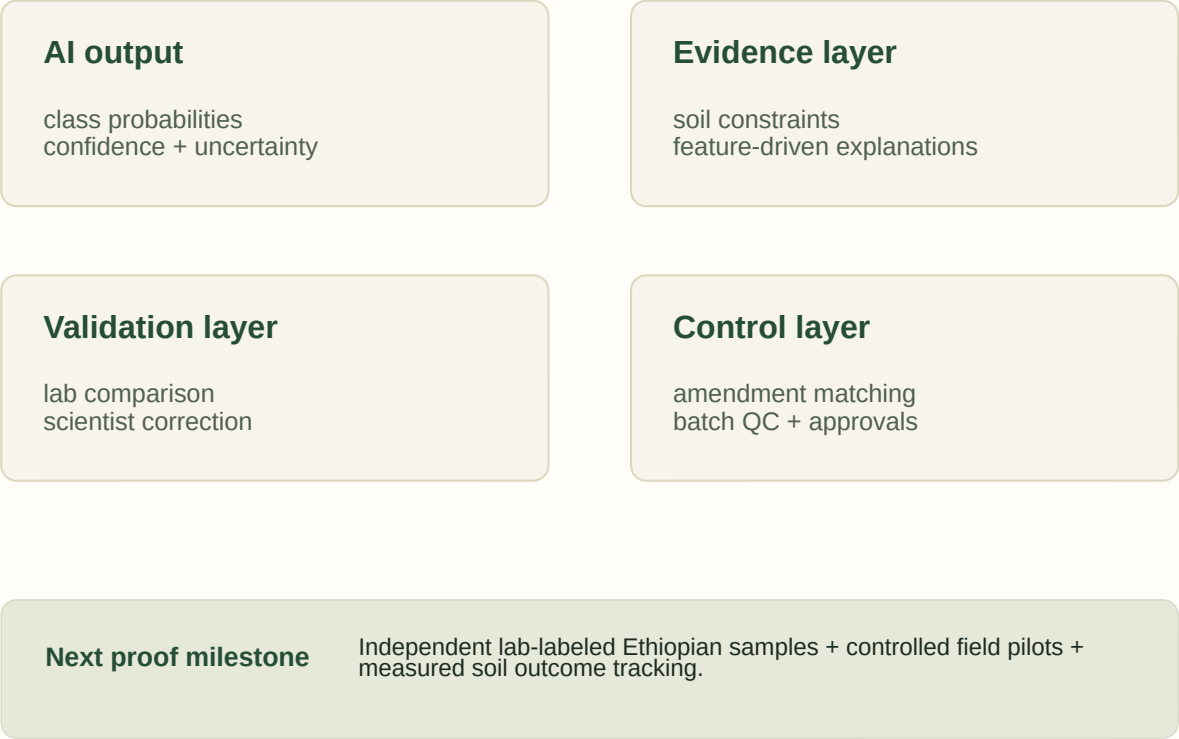


Core proposition: AI supports science; science supervises AI.



Functional prototype built; next step is independent Ethiopian validation

The right message is not “validated at national scale.” It is “built, working, scientifically guarded, and ready for structured validation.”



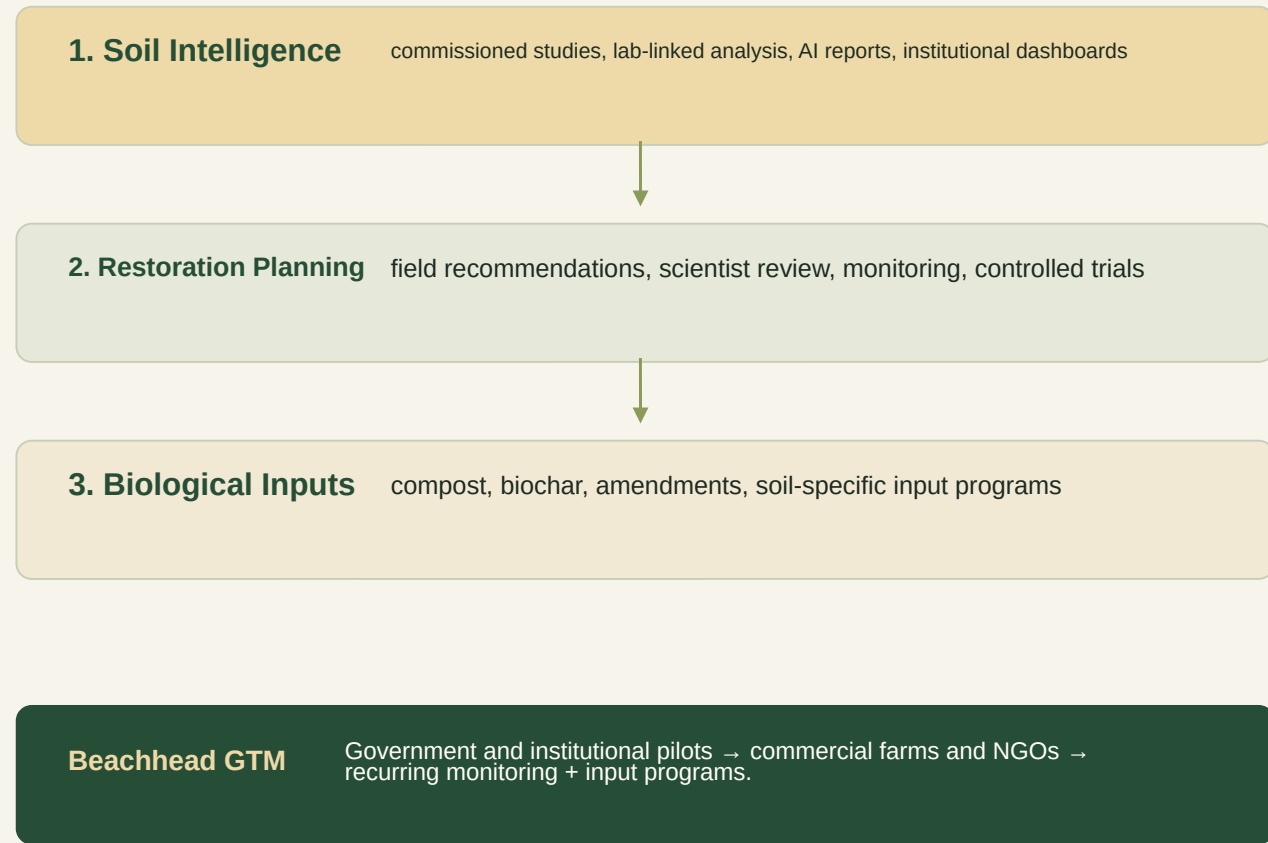
Current technical assets

- ML-backed soil status prototype
- FastAPI model service pattern
- Report metadata: model + knowledge-base versions
- Confidence, uncertainty, OOD and data-quality signals
- AI-vs-lab comparison and scientist review
- Research-controlled amendment catalog



Start with paid soil intelligence; expand into verified restoration programs

Intelligence creates the customer relationship. Controlled restoration and biological inputs create the long-term operating model.

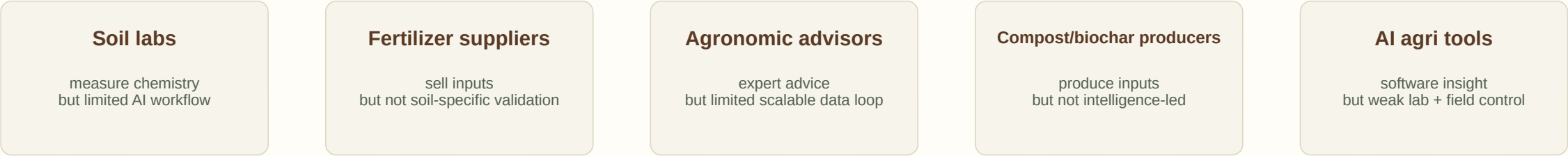


Circular layer

Organic waste becomes feedstock for compost, biochar and biological soil inputs — but only after scientific fit and controlled use are proven.

GerahtAI integrates what competitors usually separate

The defensibility is not one feature. It is the closed loop between local data, scientific validation, controlled interventions, and field outcomes.



GerahtAI differentiation

Localized Soil Intelligence + AI Confidence/Uncertainty + Lab Validation + Scientist Review + Controlled Amendment QC + Field Traceability

Why it matters

Every new validated sample and treatment outcome improves the system’s data advantage — while safe approval workflows create trust with institutions.

Use EAI to convert a working prototype into validated deployment infrastructure

The 4–6 month program should unlock data, compute, scientific validation, pilots, and investor/grant readiness.

Core team

Bisrat Seyoum — CEO / strategy
Tomas Asefa — software engineering, development, research
Advisors needed: soil science, agronomy, lab QA, waste processing

3-year planning case

Y1: pilots + paid studies
Y2: institutional contracts
Y3: recurring monitoring + input programs
Numbers remain assumption-based until pilots close.

EAI partnership ask

HPC/GPU infrastructure • AI/ML mentorship • soil dataset access • lab partnerships • controlled field pilots • investor/grant introductions

4–6 month milestones: validate with real Ethiopian samples, compare AI to lab interpretation, run controlled pilots, refine scientific reports, and prepare first institutional deployments.

Outcome **Prototype → validated pilot system → commercial readiness**

